

Ms Davis
Mrs Greenberg
Ms Lau

Mr Feng
Mrs Kerr

Name:

Teacher:.....



Pymble Ladies' College

MATHEMATICS EXTENSION 1

HSC ASSESSMENT

TASK 3

Term 2
2021

Time Allowed: 50 minutes

General Instructions

- Start each question in a **new** booklet.
- Write using black or blue pen.
- Erasable pens are not to be used.
- Approved calculators may be used.
- Diagrams are not drawn to scale.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or untidy work.
- A reference sheet is provided.

Mark	/30
Rank	/
Highest Mark	/30

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Section I

Multiple Choice. (Each question in this section is worth 1 mark)

Use Multiple Choice answer sheet for questions 1 - 6.

1 What is the value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)$?

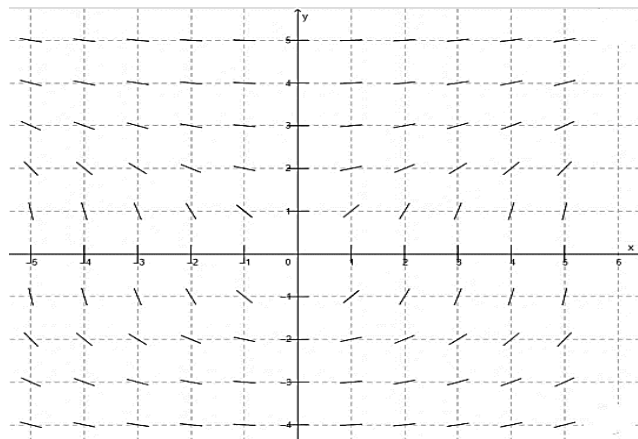
- (A) $\frac{\pi}{6}$
- (B) $\frac{5\pi}{6}$
- (C) $\frac{-\pi}{6}$
- (D) $\frac{-5\pi}{6}$

2 Which of the following is not equivalent to $\int \frac{-dx}{\sqrt{1-x^2}}$?

- (A) $\sin^{-1}(-x)$
- (B) $-\sin^{-1}(x)$
- (C) $\cos^{-1}(-x)$
- (D) $\cos^{-1}(x)$

3 Which of the given differential equations could produce the slope field shown?

- (A) $\frac{dy}{dx} = \frac{x}{y^2}$
- (B) $\frac{dy}{dx} = xy$
- (C) $\frac{dy}{dx} = \frac{x}{y}$
- (D) $\frac{dy}{dx} = x^2y$



- 4 What is $\int \frac{dx}{25+4x^2}$?
- (A) $\tan^{-1}\left(\frac{2x}{5}\right) + C$
- (B) $\frac{5}{x} \tan^{-1}\left(\frac{2x}{5}\right) + C$
- (C) $\frac{1}{25} \tan^{-1}\left(\frac{2x}{5}\right) + C$
- (D) $\frac{1}{10} \tan^{-1}\left(\frac{2x}{5}\right) + C$
- 5 Which expression is equivalent to $\int_0^{\frac{\pi}{3}} \tan x \sec^2 x dx$ when using the substitution $u = \tan x$?
- (A) $\int_0^{\frac{1}{\sqrt{3}}} u du$
- (B) $\int_0^{\sqrt{3}} u du$
- (C) $\int_0^{\frac{1}{\sqrt{3}}} u(u^2 + 1) du$
- (D) $\int_0^{\sqrt{3}} u(u^2 + 1) du$
- 6 Which of the following satisfies the differential equation $2x - y'e^y = 0$?
- (A) $y = \ln|x|$
- (B) $y = e^x$
- (C) $y = \ln|x^2 + 4|$
- (D) $y = e^{x^2}$

Section II

Answer each Question in a SEPARATE Booklet

Question 7 (8 marks)	Marks
(a) Given $y = x \sin^{-1} x$, find $\frac{dy}{dx}$.	2
(b) The area bounded by the curve $y = \sin x$, the x -axis and the line $x = \frac{\pi}{3}$ is rotated about the x -axis. Find the volume generated.	3
(c) Evaluate $\int_{-1}^0 x\sqrt{1+x} \, dx$ by using the substitution $u = 1+x$.	3

Question 8 (8 marks)

Answer in a SEPARATE Booklet

Marks

(a) Show that $y = \sin x$ satisfies the differential equation $y'' + y = 0$. **2**

(b) Find $\int \frac{\ln 2x}{x} dx$ by using the substitution $u = \ln 2x$. **2**

(c) (i) Find the x coordinate of the intersection point of the curves **1**
 $y = (x+1)^2$ and $y = (x-3)^2$.

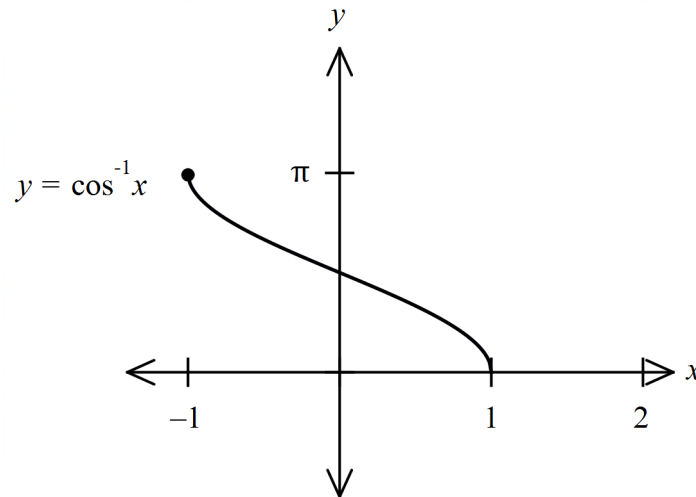
(ii) Find the volume of the solid generated when the area bounded by the curves, **3**
 $y = (x+1)^2$, $y = (x-3)^2$ and the x -axis is rotated about the x -axis.

Question 9 (8 marks)

Answer in a SEPARATE Booklet

Marks

- (a) Find the area bounded by the curve $y = \cos^{-1} x$ and the x and y axes.

3

- (b) The area $A \text{ cm}^2$ is occupied by a bacterial colony. The colony has its growth modelled by the logistic equation $\frac{dA}{dt} = \frac{1}{25} A(50 - A)$ where $t \geq 0$ and t is measured in days. At time $t = 0$, the area occupied by the bacteria colony is $\frac{1}{2} \text{ cm}^2$.

(i) Show that $\frac{1}{A(50 - A)} = \frac{1}{50} \left(\frac{1}{A} + \frac{1}{50 - A} \right)$. **1**

- (ii) Using the result from above in (i), solve the logistic equation and hence **3**

show that $A = \frac{50}{1 + 99e^{-2t}}$.

- (iii) According to this model, what is the limiting area of the bacterial colony? **1**

End of Assessment

1. C

2. C

3. A

4. D

5. B

6. C

Question 7 (8 marks)

a) $y = x \sin^{-1} x$

$$y' = 1 \times \sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \times x$$

$$= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} \quad (2)$$

b) Volume

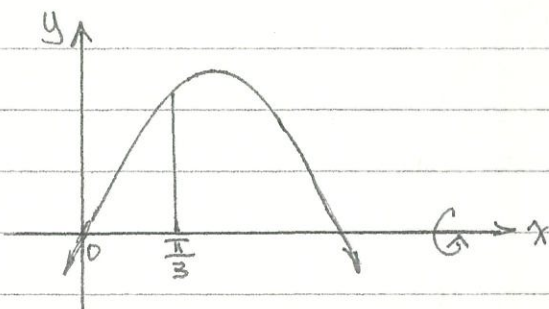
$$= \pi \int_0^{\frac{\pi}{3}} \sin^2 x \, dx$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{3}} (1 - \cos 2x) \, dx$$

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{3}}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} - 0 + \frac{1}{2} \sin 0 \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \text{ cubic units} \quad (3)$$



c) $\int_{-1}^0 x \sqrt{1+x} \, dx$

$$u = 1+x$$

$$= \int_0^1 \sqrt{u} (u-1) \, du$$

$$du = dx$$

$$x=0, u=1+0=1$$

$$x=-1, u=1+(-1)=0$$

$$= \int_0^1 (u^{\frac{3}{2}} - u^{\frac{1}{2}}) \, du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= \frac{2}{5} - \frac{2}{3} - 0 + 0$$

$$= \frac{-4}{15} \quad (3)$$

Question 8 (8 marks)

a) $y = \sin x$

$y' = \cos x$

$y'' = -\sin x$

LHS = $y'' + y$

$= -\sin x + \sin x$

$= 0$

$= \text{RHS}$

(1)

(2)

b) $\int \frac{\ln 2x}{x} dx$

$u = \ln 2x$

$= \int u du$

$du = \frac{1}{x} dx$

$= \frac{1}{2} u^2 + C$

$= \frac{1}{2} (\ln 2x)^2 + C$

(1)

(2)

c) i) $(x+1)^2 = (x-3)^2$

$x^2 + 2x + 1 = x^2 - 6x + 9$

$x = 1$

(1)

ii) Volume

$= \pi \int_{-1}^1 (x+1)^4 dx$

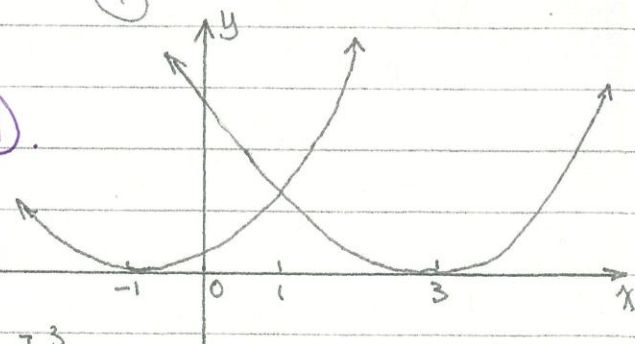
$+ \pi \int_1^3 (x-3)^4 dx$

$= \pi \left[\frac{1}{5} (x+1)^5 \right]_{-1}^1$

$+ \pi \left[\frac{1}{5} (x-3)^5 \right]_1^3$

$= \pi \left[\frac{1}{5} (2)^5 - 0 + 0 - \frac{1}{5} (-2)^5 \right]$

$= \frac{64\pi}{5} \text{ cubic units}$



(3)

Question 9 (8 marks)

a) $y = \cos^{-1} x$

$\Rightarrow x = \cos y$

Area $\frac{\pi}{2}$

$= \int_0^{\frac{\pi}{2}} \cos y \, dy$

$= [\sin y]_0^{\frac{\pi}{2}}$

$= \sin \frac{\pi}{2} - \sin 0$

$= 1 - 0$

$= 1 \text{ square unit}$

③ — Correct

② — Correct integral, correct limits, correct integration

① — correct integral

$\int_0^{\frac{\pi}{2}} \cos x \, dx \rightarrow \textcircled{1}$

③

b) i) $RHS = \frac{1}{50} \left(\frac{1}{A} + \frac{1}{50-A} \right)$

$= \frac{1}{50} \times \frac{50-A+A}{A(50-A)}$

$= \frac{1}{50} \times \frac{50}{A(50-A)}$

$= \frac{1}{A(50-A)}$

$= LHS$

① r/w

ii) $\frac{dA}{dt} = \frac{1}{25} A(50-A)$

$\int \frac{dA}{A(50-A)} = \int \frac{dt}{25}$

$\frac{1}{50} \int \left(\frac{1}{A} + \frac{1}{50-A} \right) dA = \int \frac{dt}{25}$

$\ln A - \ln(50-A) = 2t + C$

$\ln \left(\frac{50-A}{A} \times k \right) = -2t$

③ — All correct

When $t = 0$, $A = \frac{1}{2}$

$\ln \left(\frac{50-0.5}{0.5} k \right) = 0$

$99k = 1$

$k = \frac{1}{99}$

$\ln \left(\frac{50-A}{A} \times \frac{1}{99} \right) = -2t$

② — Correctly integrate with initial condition

① — Just the set

up

$$\left(\frac{50-A}{A} \times \frac{1}{99} = e^{-2t} \right.$$

$$\left(\frac{50-A}{A} = 99e^{-2t} \right.$$

$$50-A = 99Ae^{-2t}$$

$$50 = A + 99Ae^{-2t}$$

$$50 = A(1 + 99e^{-2t})$$

$$A = \frac{50}{1 + 99e^{-2t}} \quad (3)$$

iii) As $t \rightarrow \infty$,

$$\left(e^{-2t} \rightarrow 0, 1 + 99e^{-2t} \rightarrow 1 \right.$$

$$\text{so } A \rightarrow 50$$

$$\therefore \text{Limiting area} = 50 \text{ cm}^2 \quad (1)$$